

DQC-SI: Distributed Quantum Computing Systems and Infrastructure Workshop

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I. WORKSHOP TITLE

DQC-SI: Distributed Quantum Computing Systems and Infrastructure

II. WORKSHOP SUMMARY

This workshop explores systems and infrastructure for Distributed Quantum Computing (DQC), with an emphasis on how coordinated design enables scalable multi-QPU execution. Attendees will gain insights into integrating physical, network, system, and application layers to improve scalability, reliability, and overall efficiency in DQC.

III. ABSTRACT

As quantum computing systems continue to scale, DQC has emerged as a promising paradigm to overcome the limitations of individual quantum processors [1, 2]. In DQC, multiple quantum processing units (QPUs) are interconnected through quantum networks, enabling them to collaboratively execute computational tasks that exceed the capabilities of a single device [3, 4]. This shift from isolated processors to interconnected systems represents a fundamental step toward scalable quantum computing. However, **current approaches largely treat system layers in isolation**, separating physical-layer constraints [5, 6], network protocols [7, 8], and application-level execution [9, 10], which leads to inefficiencies in scalability and reliability. As quantum systems grow in size and complexity, these limitations become increasingly significant. **This workshop focuses on system and infrastructure design for DQC**, highlighting coordinated design across physical, network, system, and application layers. By explicitly connecting these layers, cross-layer approaches can capture the interplay between hardware limitations, communication processes, and application requirements, enabling more efficient and scalable system designs. This perspective also opens new opportunities for co-design across traditionally separate research areas. By bringing together researchers from quantum computing, networking, systems, and applications, this workshop

aims to establish cross-layer infrastructure as a foundational principle for scalable quantum systems. The long-term vision is to enable efficient quantum data centers and networked quantum systems that support large-scale scientific and industrial applications.

IV. ORGANIZERS

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V. OBJECTIVES

The short-term objective of this workshop is to introduce the key challenges and emerging solutions in cross-layer infrastructure for DQC. The workshop will highlight how physical-layer constraints, quantum networking processes, and system-level execution are tightly interconnected, and why treating them in isolation limits scalability and efficiency. Through invited talks and discussions, the workshop will present state-of-the-art approaches in DQC cross-layer infrastructure, while fostering interdisciplinary exchange among researchers in quantum computing, networking, systems, and applications.

The long-term vision is to establish cross-layer infrastructure as a foundational paradigm for scalable quantum systems and to build a sustained research community around this theme. The workshop aims to catalyze the development of a shared research roadmap and open challenges for DQC infrastructure. It also seeks to promote common frameworks, benchmarks, and simulation platforms that integrate multiple system layers, ultimately supporting the realization of scalable quantum systems for scientific discoveries

and industrial applications. To support this vision, we will initiate a structured community-building effort, including the formation of a DQC working group within the broader quantum research community, as well as the coordination of follow-up activities such as special sessions, collaborative projects, and future workshops.

VI. RELEVANCE

This workshop aligns with several QCE26 topics, including Distributed Quantum Computing, Quantum Systems Engineering, and Quantum Communications & Cryptography. It focuses on the infrastructure required to enable scalable quantum systems composed of interconnected QPUs, covering areas such as quantum interconnects, distributed execution, system software, and hardware-software co-design.

Unlike existing efforts that focus on individual layers, this workshop emphasizes cross-layer integration across physical, network, and system levels, bridging gaps between quantum networking, distributed computing, and software infrastructure. This perspective supports the development of scalable, end-to-end quantum systems and contributes to advancing practical quantum computing and future DQC architectures.

VII. ANTICIPATED OUTCOMES

The workshop is expected to catalyze a cross-layer DQC infrastructure community that connects researchers in quantum computing, networking, systems, and applications. It will identify key open challenges and foster collaboration across these traditionally separate areas, generating synergy between quantum networking and distributed quantum computing. A key outcome will be the development of an initial research roadmap for cross-layer quantum infrastructure, including core design principles, system abstractions, and priority research directions. The workshop will also encourage progress toward shared frameworks and benchmarks.

VIII. WORKSHOP FORMAT

The workshop will be interactive, combining invited talks, contributed presentations, and discussions. We plan to invite 1 keynote and 2-3 invited speakers from academia, industry, or national laboratories to provide diverse perspectives on DQC and cross-layer system design. We will solicit workshop papers through peer review. Accepted papers will be presented at the workshop.

Tentative Agenda:

- Session 1: Physical- and Link-Layer Foundations for DQC
- Session 2: Quantum Networking and Resource Management in DQC

- Session 3: Systems and Software for DQC Execution
- Session 4: DQC Applications and Cross-Layer Co-Design
- Discussion on DQC Design Challenges and Future Directions
- Closing Remarks

IX. WORKSHOP PAPERS

We will solicit papers on topics related to cross-layer infrastructure for DQC. Submissions may include original research, system designs, early-stage ideas, or position papers. Tentative CFP below.

Call for Papers: Distributed Quantum Computing Systems and Infrastructure (DQC-SI) Workshop

As quantum computing systems continue to scale, Distributed Quantum Computing (DQC) has emerged as a promising paradigm to overcome the limitations of individual quantum processors. In DQC, multiple quantum processing units (QPUs) are interconnected through quantum networks, enabling them to collaboratively execute computational tasks that exceed the capabilities of a single device. However, current research remains largely siloed across system layers, separating physical-layer constraints, networking protocols, and system-level execution, which limits scalability, efficiency, and reliability. The DQC-SI workshop focuses on **system and infrastructure design for DQC**, where coordinated design across physical, network, and system layers is essential. This workshop aims to provide an open platform for researchers from quantum computing, networking, and systems to present state-of-the-art developments, exchange ideas, and foster collaboration. By bridging these separate domains, DQC-SI seeks to advance scalable, end-to-end quantum system design and support the development of future quantum data centers and networked quantum platforms.

Topic of Interests

We invite submissions to the DQC-SI workshop to be held in conjunction with IEEE Quantum Week (QCE 2026). The workshop will provide a forum for researchers and practitioners to present and discuss recent advances in distributed quantum computing infrastructure and cross-layer system design.

Topics of interest include, but are not limited to:

- System and infrastructure design for DQC
- Network-aware quantum circuit partitioning
- Distributed circuit execution and compilation
- Quantum interconnects and network topologies
- Quantum systems software, compilers, and middleware for distributed execution
- Quantum networking, entanglement routing, resource management for DQC
- Cross-layer DQC simulation and benchmarking
- Hybrid quantum-classical distributed systems
- Applications enabled by DQC

Submission Guidelines

We welcome submissions of original research papers, work-in-progress papers, and position papers related to distributed quantum computing infrastructure and cross-layer system design. Submissions should follow the IEEE conference format and must not exceed **4 pages**, including references. All submissions will be peer-reviewed by the workshop program committee. DQC-SI workshop papers will be included in the QCE'26 Proceedings. Submissions must follow the IEEE formatting guidelines, with a 24pt title and 10pt main text.

L^AT_EX users should use `\documentclass[10pt, conference]{IEEEtran}` without the `comsoc` or `comsocconf` options. All submissions must comply with the IEEE Policy on Authorship.

The review process will be single-blind; therefore, author names and affiliations should be included in the submission. Papers are limited to **four pages**, including references.

Important Dates:

- Submission Deadline: TBD
- Notification of Acceptance: TBD
- Camera-Ready Deadline: TBD
- Workshop Date: QCE 2026

Tentative Program Committee:

- Zhen Ni, Florida Atlantic University
- Ruozhou Yu, NC State University
- Chen Qian, University of California Santa Cruz
- Walid Saad, Virginia Tech
- Brian Mark, George Mason University

Main Contact: Zebo Yang, yangz@fau.edu

For more information, please contact the workshop organizers or visit the workshop webpage (to be announced). We look forward to your submissions!

X. TARGET AUDIENCE

This workshop targets researchers and practitioners in quantum computing, quantum networking, computer systems, and applications, particularly those interested in scalable and distributed quantum infrastructures. It is also relevant to researchers in computer architecture, distributed systems, and high-performance computing who are exploring the integration of quantum technologies into larger computing ecosystems. The workshop is designed to be accessible to participants with a general background in quantum computing or computer systems, with an introductory context provided to ensure a common understanding across disciplines. Attendees are expected to have a basic understanding of quantum computing concepts, while expertise in specific subfields (e.g., networking or systems) is not required. The workshop aims to attract a diverse mix of participants from academia, industry, and national laboratories, including those working on quantum hardware, software, and applications. By

bringing together these communities, the workshop fosters interdisciplinary collaboration and knowledge exchange across traditionally separate areas.

XI. WORKSHOP PARTICIPANTS

The expected minimum and maximum number of attendees are 20 and 50, respectively. We plan to include 1 keynote speaker and 2–3 invited speakers, along with a research presentation track consisting of peer-reviewed paper submissions. Participants will be solicited through multiple channels, including the QCE website and mailing lists, as well as announcements on professional platforms such as LinkedIn and other academic networks.

XII. WORKSHOP ORGANIZER BIOGRAPHIES

Zebo Yang (<https://zeboyang.com/>): Dr. Yang is an Assistant Professor in the Department of Electrical Engineering and Computer Science at Florida Atlantic University. He received his Ph.D. in Computer Science from Washington University in St. Louis in May 2025. Prior to this, he worked as a Senior Software Engineer at companies such as Tencent, Baidu, and DJI from 2011 to 2017. He received his master's degree in computer engineering from Waseda University, Tokyo, Japan, in 2019. His research interests include quantum computing, quantum networking, distributed quantum computing, quantum optimization, and machine learning.

Travis S Humble (<https://www.ornl.gov/staff-profile/travis-s-humble>): Dr. Humble is the director of the Quantum Science Center, a Distinguished Scientist at Oak Ridge National Laboratory, and the director of the lab's Quantum Computing Institute. Travis is leading the development of new quantum technologies and infrastructure to impact the DOE mission of scientific discovery through quantum computing. As the director of the QSC, Travis leads the innovation of scalable, resilient quantum information technologies through new materials, devices, and algorithms, and facilitates the transfer of quantum technologies to the broadest audience. Travis serves as the director of the OLCF Quantum Computing User Program by leading the management and operation of quantum computing technologies for a broad base of users. These revolutionary new approaches to familiar computational problems help reduce algorithmic complexity, reduce computational resource requirements like power and communication, and increase the scale at which state-of-the-art scientific applications perform. In this role, Travis leads the design, development, and benchmarking of quantum computing platforms. Travis is the editor-in-chief for ACM Transactions on Quantum Computing, Associate Editor for Quantum Information Processing, and co-chair of the IEEE Quantum Initiative. Travis also holds a joint faculty appointment

with the University of Tennessee Bredesen Center for Interdisciplinary Research and Graduate Education working with students on energy-efficient computing solutions. Travis received a doctorate in theoretical chemistry from the University of Oregon before joining ORNL in 2005.

Vicente A Leyton Ortega (<https://www.ornl.gov/staff-profile/vicente-leyton-ortega>): Dr. Leyton is a research scientist in quantum information science at Oak Ridge National Laboratory, where he works within the Quantum Computational Science Group. He earned his Ph.D. in Physics from the University of Hamburg in 2012. Prior to joining ORNL in 2019, he worked as a consultant for Rigetti Computing and served as an Associate Professor at Universidad Santiago de Cali from 2015 to 2017. He was a postdoctoral researcher at Universidad Antonio Nariño. His research focuses on quantum computation, error mitigation, hybrid quantum-classical algorithms, and quantum machine learning. He has contributed to the development of practical methods for near-term quantum systems and has co-organized multiple workshops in the area of quantum information science.

Weiwen Jiang (<https://jqub.ece.gmu.edu/>): Dr. Jiang is an Assistant Professor at George Mason University. He is the associate director and incoming co-director of Quantum Science & Engineering Center at Mason. He was a Post-Doctoral Research Associate at the University of Notre Dame. He received his Ph.D. degree from Chongqing University in 2019. From 2017 to 2019, he was a research scholar in the Department of Electrical and Computer Engineering at the University of Pittsburgh. His research interest is in hardware and software co-design, in particular, the co-design of neural networks and different hardware accelerators, including mobile devices, FPGA, ASIC, and quantum computing. Dr. Jiang's research works have been published in prestigious journals and conferences, including Nature Electronics, Nature Communications, IEEE/ACM Transactions, DAC, ICCAD, QuantumWeek, ESWEK, GLSVLSI, etc. He is the recipient of the Best Paper Award in IEEE TCAD'21 and ICCD'17. His work also obtained the Best Paper Nominations in DAC'19, CODES+ISSS'19, and ASP-DAC'20. He is the QEC track co-chair from 2023 to 2025. The funding organizer of the StableQ workshop.

Lei Yang (<https://leiyang0416.github.io/>): Dr. Yang is currently an Assistant Professor in the Department of Information Sciences and Technology at George Mason University. Before that, she was an Assistant Professor in the Department of Electrical and Computer Engineering at the University of New Mexico, USA. She received her Ph.D. degree and B.E. degree in 2019 and 2013, respectively, from Chongqing University, China. She was a Post-Doctoral Research Associate in the Department of Computer Science

and Engineering at the University of Notre Dame. Her primary research interests lie in the joint area of Hardware/Software Co-Exploration for Neural Network Architectures, Network-on-Chip, and Quantum Networking.

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